DOCUMENTACIÓN HADOOP

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Índice

1. Instalamos ssh.

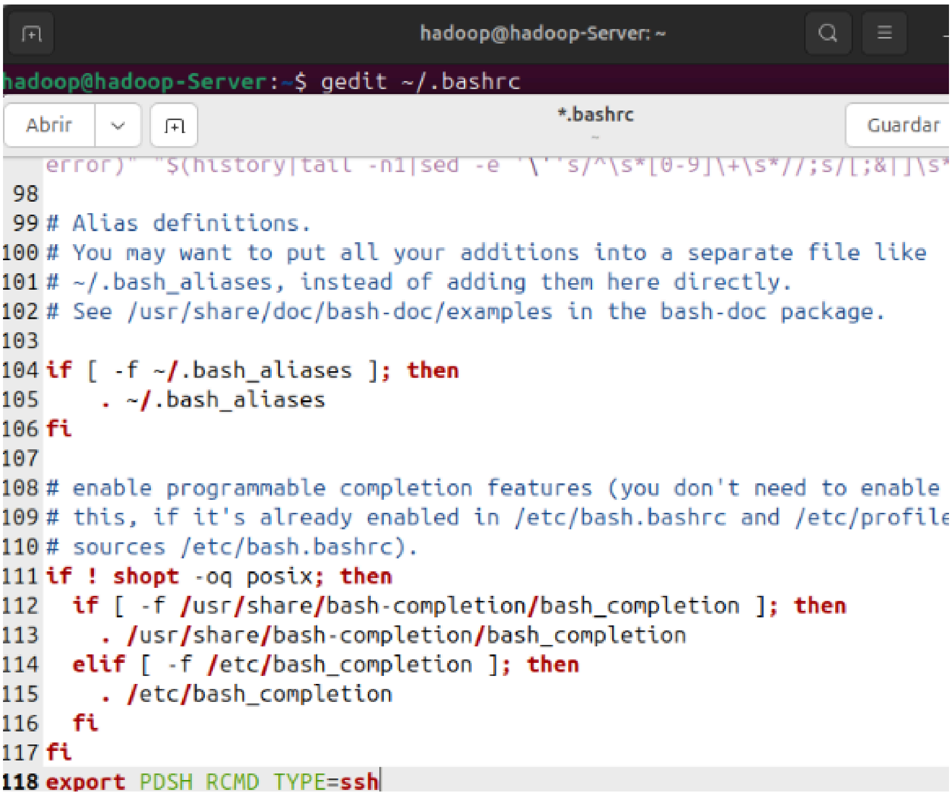
```
hadoop@hadoop-Server: ~
hadoop@hadoop-Server:~$ sudo apt install ssh
[sudo] contraseña para hadoop:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no
son necesarios.
  libgl1-amber-dri libglapi-mesa libllvm17t64 python3-netifaces
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes adicionales:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
Paquetes sugeridos:
  molly-guard monkeysphere ssh-askpass
Se instalarán los siguientes paquetes NUEVOS:
  ncurses-term openssh-server openssh-sftp-server ssh ssh-import-id
0 actualizados, 5 nuevos se instalarán, 0 para eliminar y 2 no actualizados.
Se necesita descargar 837 kB de archivos.
Se utilizarán 6.801 kB de espacio de disco adicional después de esta operación.
¿Desea continuar? [S/n]
Des:1 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-
server amd64 1:9.6p1-3ubuntu13.14 [37,3 kB]
Des:2 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-serve
r amd64 1:9.6p1-3ubuntu13.14 [510 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+
```

2. Instalamos pdsh.

```
hadoop@hadoop-Server:~$ sudo apt install pdsh
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no
son necesarios.
  libgl1-amber-dri libglapi-mesa libllvm17t64 python3-netifaces
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes adicionales:
  genders libgenders0 ssh-askpass
Paquetes sugeridos:
  rdist
Se instalarán los siguientes paquetes NUEVOS:
  genders libgenders0 pdsh ssh-askpass
0 actualizados, 4 nuevos se instalarán, 0 para eliminar y 2 no actualizados.
Se necesita descargar 201 kB de archivos.
Se utilizarán 630 kB de espacio de disco adicional después de esta operación.
¿Desea continuar? [S/n]
Des:1 http://es.archive.ubuntu.com/ubuntu noble/universe amd64 libgenders0 amd64
 1.22-1build8 [30,8 kB]
Des:2 http://es.archive.ubuntu.com/ubuntu noble/universe amd64 genders amd64 1.2
2-1build8 [31,0 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu noble/universe amd64 ssh-askpass amd64
```

3. Abrimos el archivo .bashrc y añadimos al final del archivo:

export PDSH_RCMD_TYPE=ssh



4. Creamos una nueva key para el ssh.

```
hadoop@hadoop-Server:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoop/.ssh/id_rsa):
Your identification has been saved in /home/hadoop/.ssh/id_rsa
Your public key has been saved in /home/hadoop/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:mxN1s1cGJhH9HcpRpgPo0k46Or00YkxZjCeNLIesRJg hadoop@hadoop-Server
The key's randomart image is:
+---[RSA 3072]----+
|.o         ..+++o |
|E . o =   .  .++o |
| . + * =o . +oo.=|
|. . o =. = . =.oo|
| .    o  S   . .  |
|    o  o =    .   |
|     +oo=         |
|    .oo...        |
|      ...         |
+----[SHA256]-----+
```

5. Copiamos la nueva key al archivo authorized_keys.

```
hadoop@hadoop-Server:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

6. Verificamos la configuración de SSH.

```
hadoop@hadoop-Server:~$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:RKW+OY1+QSCDeOImdnbStM25Ss9kLLr9UoE82K5Ex5Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

El mantenimiento de seguridad expandido para Applications está desactivado

Se puede aplicar 1 actualización de forma inmediata.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»


1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
```

7. Instalamos Java 8.

```
hadoop@hadoop-Server:~$ sudo apt install openjdk-8-jdk
[sudo] contraseña para hadoop:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y y
son necesarios.
  libgl1-amber-dri libglapi-mesa libllvm17t64 python3-netifaces
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes adicionales:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11proto-dev
  xorg-sgml-doctools xtrans-dev
Paquetes sugeridos:
  default-jre libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc
  openjdk-8-demo openjdk-8-source visualvm fonts-nanum fonts-ipafont-gothic
  fonts-ipafont-mincho fonts-wqy-microhei fonts-wqy-zenhei fonts-indic
Se instalarán los siguientes paquetes NUEVOS:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-8-jdk
  openjdk-8-jdk-headless openjdk-8-jre openjdk-8-jre-headless x11proto-dev
```

8. Comprobamos si Java está instalado.

```
hadoop@hadoop-Server:~$ java -version
openjdk version "1.8.0_462"
OpenJDK Runtime Environment (build 1.8.0_462-8u462-ga~us1-0ubuntu2~24.04.2-b08)
OpenJDK 64-Bit Server VM (build 25.462-b08, mixed mode)
```

9. Descargamos Hadoop.

```
hadoop@hadoop-Server:~$ sudo wget -P ~ https://mirrors.sonic.net/apache/hadoop/c
ommon/hadoop-3.4.2/hadoop-3.4.2.tar.gz
--2025-10-30 20:00:56--  https://mirrors.sonic.net/apache/hadoop/common/hadoop-3
.4.2/hadoop-3.4.2.tar.gz
Resolviendo mirrors.sonic.net (mirrors.sonic.net)... 157.131.224.201, 2001:5a8:6
01:4003:157:131:224:201
Conectando con mirrors.sonic.net (mirrors.sonic.net)[157.131.224.201]:443... con
ectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 1065831750 (1016M) [application/x-gzip]
Guardando como: '/home/hadoop/hadoop-3.4.2.tar.gz'

hadoop-3.4.2.tar.gz    0%[                    ] 515,54K   142KB/s    eta 2h 5m
```

10. Descomprimimos el archivo instalado.

```
hadoop@hadoop-Server:~$ tar xzf hadoop-3.4.2.tar.gz
```

11. Cambiamos el nombre de la carpeta de hadoop-3.4.2 a hadoop.

```
hadoop@hadoop-Server:~$ mv hadoop-3.4.2 hadoop
```

12. Abrimos el archivo hadoop-env.sh y cambiamos la línea export JAVA_HOME por:

    export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/

```
hadoop@hadoop-Server: ~
hadoop@hadoop-Server:~$ gedit hadoop/etc/hadoop/hadoop-env.sh
Abrir   *hadoop-env.sh ~/hadoop/etc/hadoop
54 # export JAVA_HOME=
55 export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/
56
```

13. Movemos la carpeta hadoop al directorio /usr/local/hadoop

```
hadoop@hadoop-Server:~$ sudo mv hadoop /usr/local/hadoop
```

14. Abrimos el archivo /etc/environment y agregamos las siguientes 2 líneas:
    a. PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/local/hadoop/bin:/usr/local/hadoop/sbin"
    b. JAVA_HOME="/usr/lib/jvm/java-8-openjdk-amd64/jre"

```
hadoop@hadoop-Server:~$ sudo gedit /etc/environment
Abrir   *environment /etc   Guardar
1 PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/local/hadoop/bin:/usr/local/hadoop/sbin"
2 JAVA_HOME="/usr/lib/jvm/java-8-openjdk-amd64/jre"
```

15. Creamos el usuario hadoopuser.

```
hadoop@hadoop-server: ~
hadoop@hadoop-server:~$ sudo adduser hadoopuser
[sudo] contraseña para hadoop:
info: Añadiendo el usuario `hadoopuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Añadiendo el nuevo grupo `hadoopuser' (1001) ...
info: Adding new user `hadoopuser' (1001) with group `hadoopuser (1001)' ...
info: Creando el directorio personal `/home/hadoopuser' ...
info: Copiando los ficheros desde `/etc/skel' ...
Nueva contraseña:
CONTRASEÑA INCORRECTA: La contraseña tiene menos de 8 caracteres
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para hadoopuser
Introduzca el nuevo valor, o presione INTRO para el predeterminado
        Nombre completo []: hadoop
        Número de habitación []: 0
        Teléfono del trabajo []: 0
        Teléfono de casa []: 0
        Otro []: 0
¿Es correcta la información? [S/n]
info: Adding new user `hadoopuser' to supplemental / extra groups `users' ...
info: Añadiendo al usuario `hadoopuser' al grupo `users' ...
```

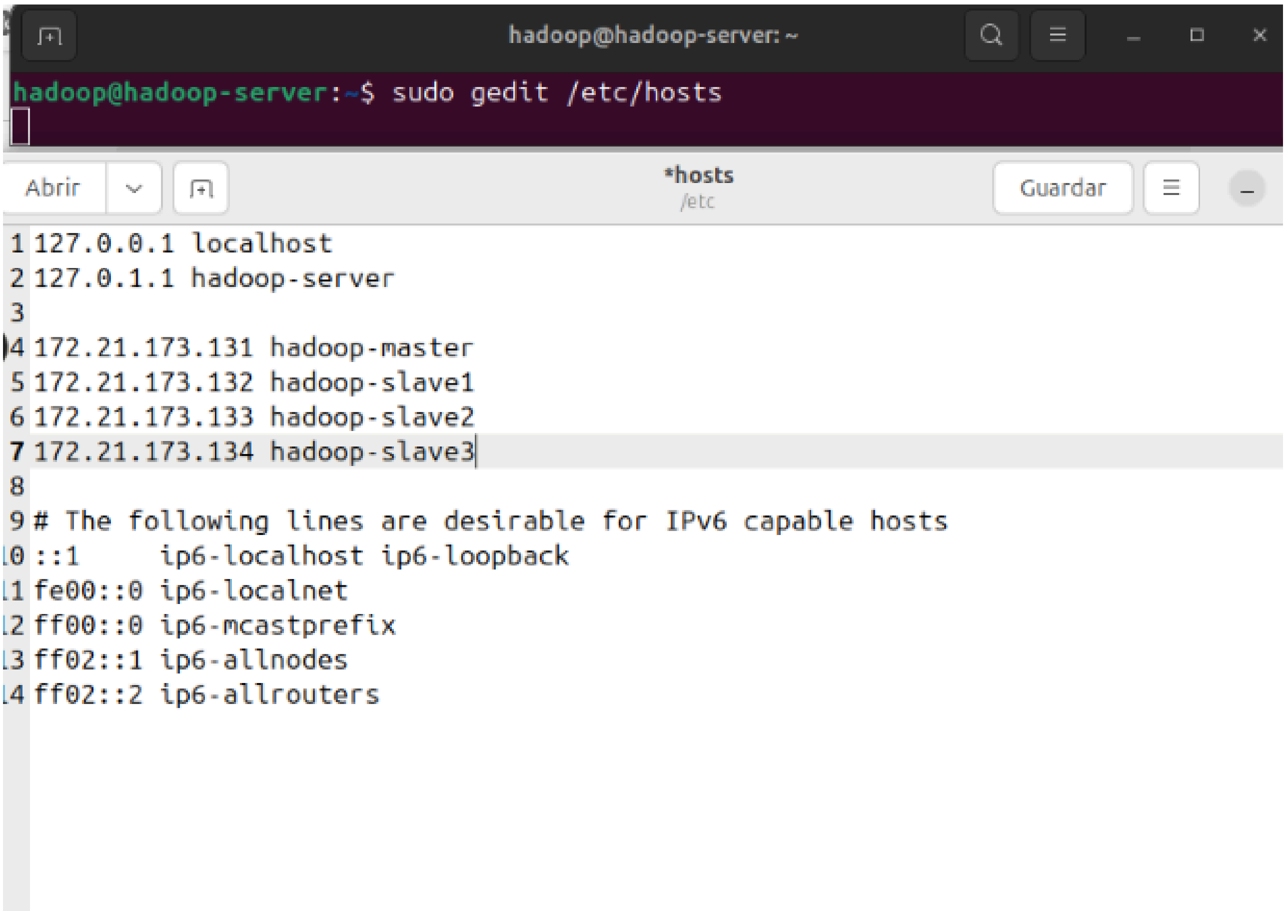
16. Cambiamos los permisos del usuario hadoopuser.

```
hadoop@hadoop-server:~$ sudo usermod -aG hadoopuser hadoopuser
[sudo] contraseña para hadoop:
hadoop@hadoop-server:~$ sudo chown hadoopuser:root -R /usr/local/hadoop/
hadoop@hadoop-server:~$ sudo chmod g+rwx -R /usr/local/hadoop/
hadoop@hadoop-server:~$ sudo adduser hadoopuser sudo
info: Añadiendo al usuario `hadoopuser' al grupo `sudo' ...
```

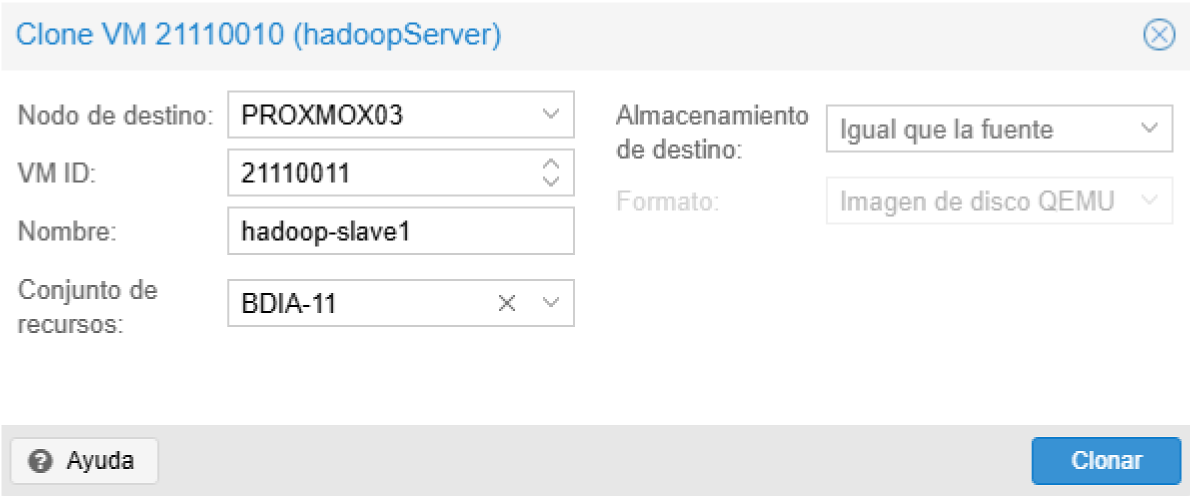
17. Verificamos la IP del ordenador.

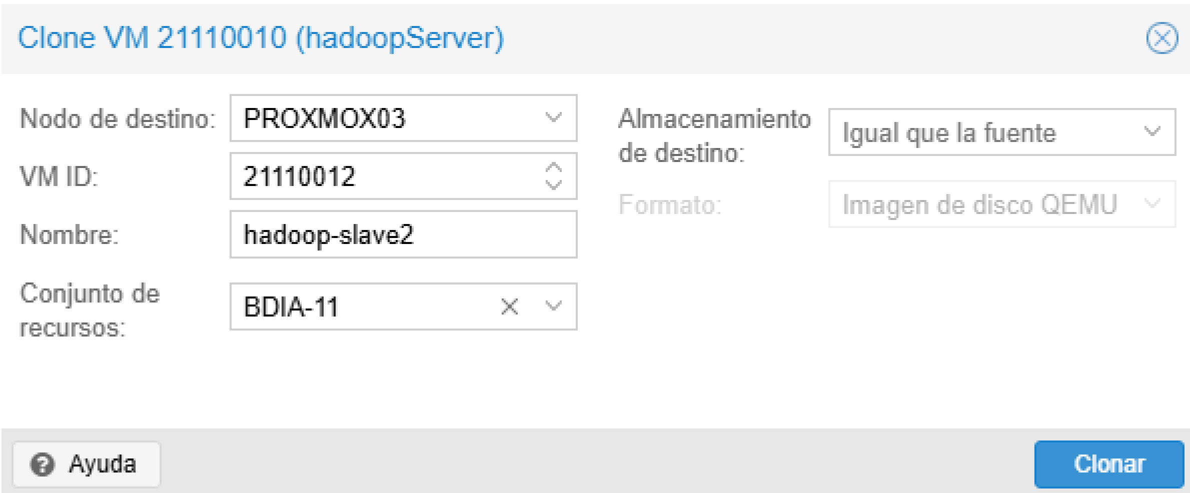
```
hadoop@hadoop-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether bc:24:11:87:c9:b3 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 172.21.173.131/26 brd 172.21.173.191 scope global dynamic noprefixroute
 ens18
       valid_lft 5634sec preferred_lft 5634sec
    inet6 fe80::be24:11ff:fe87:c9b3/64 scope link
       valid_lft forever preferred_lft forever
```

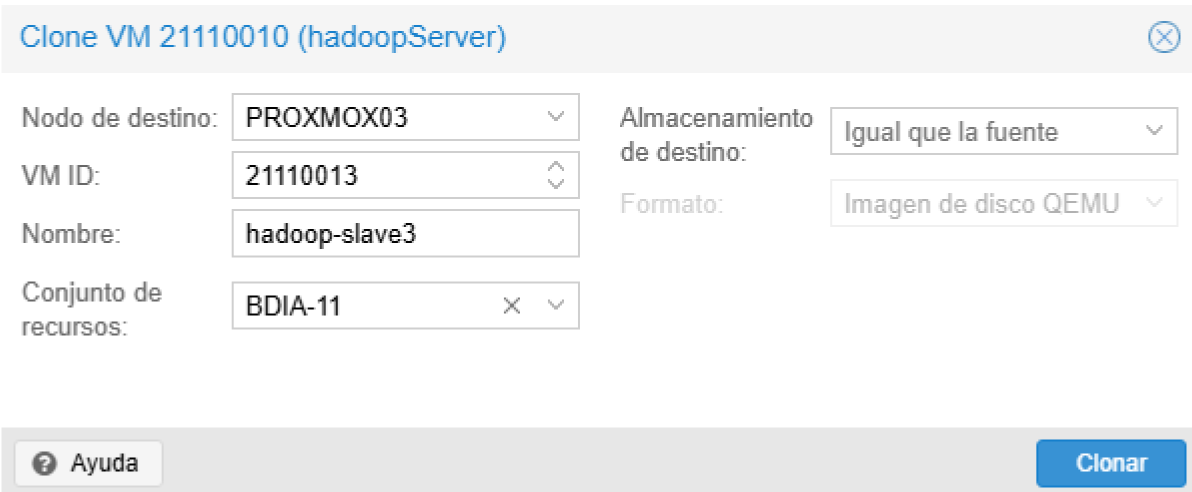
18. Abrimos el archivo /etc/hosts y agregamos las IP del master y los slaves. En mi caso son:
    a. Master: 172.21.173.131
    b. Slave1: 172.21.173.132
    c. Slave2: 172.21.173.133
    d. Slave3: 172.21.173.134



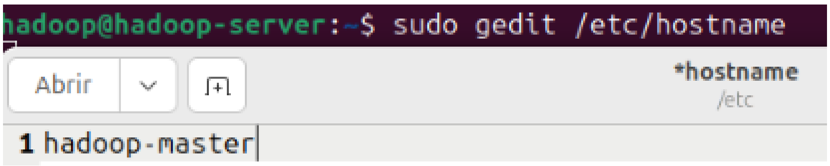
19. Creamos los slaves clonando la máquina master 3 veces.



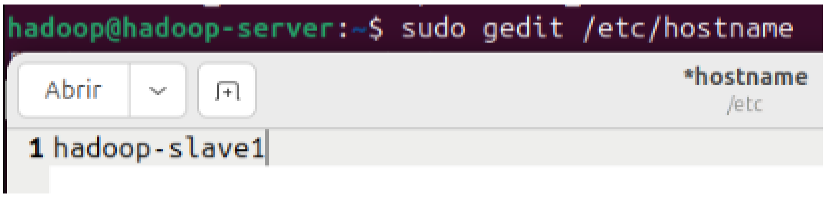


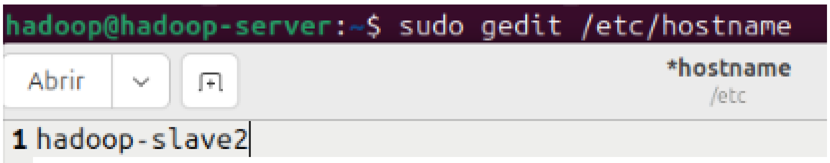


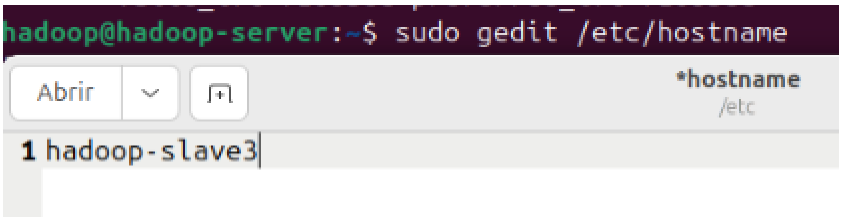
20. En el master, abrimos el archivo /etc/hostname y ponemos el nombre de nuestra máquina.



21. Hacemos lo mismo con los slaves.







22. Cambiamos de usuario a hadoopuser y creamos una key SSH.

```
hadoopuser@hadoop-master: /home/hadoop
adoop@hadoop-master:~$ su hadoopuser
ontraseña:
o run a command as administrator (user "root"), use "sudo <command>".
ee "man sudo_root" for details.

adoopuser@hadoop-master:/home/hadoop$ ssh-keygen -t rsa
enerating public/private rsa key pair.
nter file in which to save the key (/home/hadoopuser/.ssh/id_rsa):
reated directory '/home/hadoopuser/.ssh'.
nter passphrase (empty for no passphrase):
nter same passphrase again:
our identification has been saved in /home/hadoopuser/.ssh/id_rsa
our public key has been saved in /home/hadoopuser/.ssh/id_rsa.pub
he key fingerprint is:
HA256:HpRqFHATmld/nRq/aWMg8qCaEzIglULWx6YYoMkS2wI hadoopuser@hadoop-master
he key's randomart image is:
---[RSA 3072]----+
Eo..o.=..        |
**o. B + o    . .|
*++ * o o . o o  |
+o . o o    . +  |
..    o S . o .  |
  o .. o = . . o |
  o . . . .    * |
```

23. Copiamos la key SSH a todos los usuarios.

```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-master
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/
.ssh/id_rsa.pub"
The authenticity of host 'hadoop-master (172.21.173.131)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kSO/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-master's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'hadoopuser@hadoop-master'"
and check to make sure that only the key(s) you wanted were added.
```

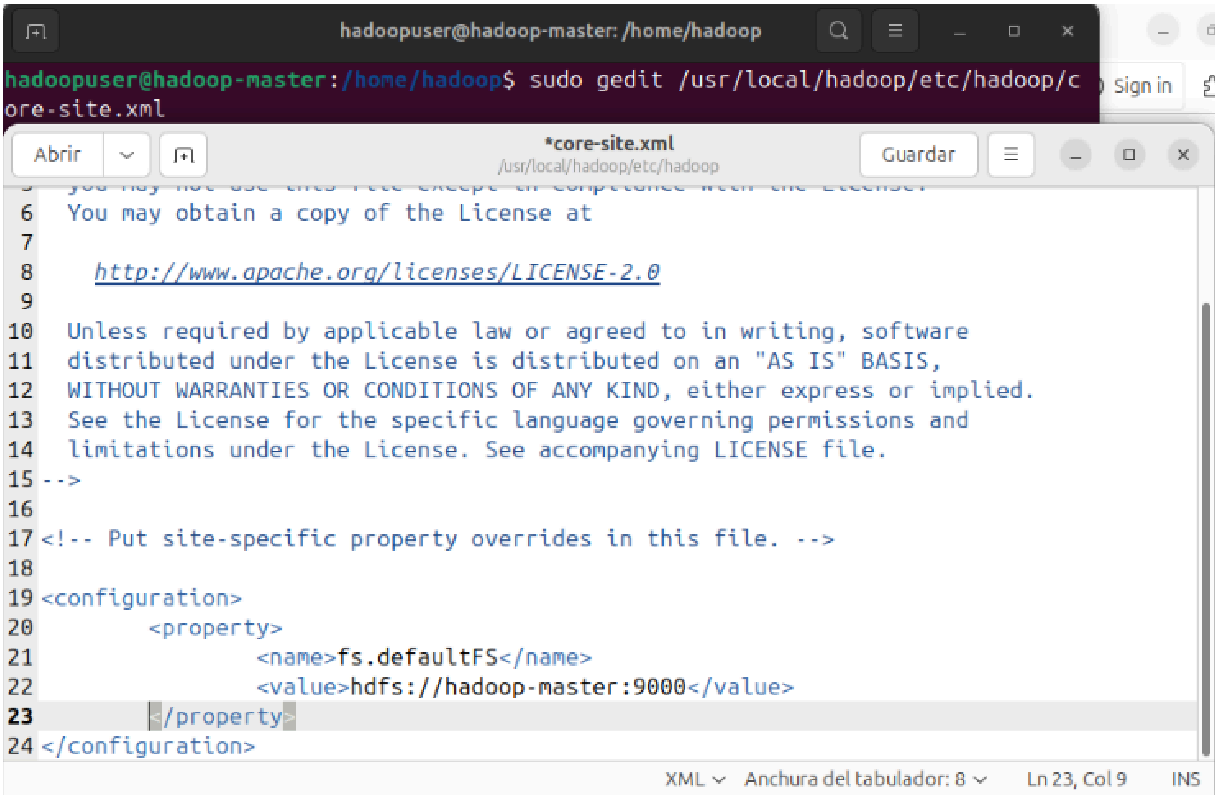
```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-slave1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/
.ssh/id_rsa.pub"
The authenticity of host 'hadoop-slave1 (172.21.173.132)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kSO/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-slave1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'hadoopuser@hadoop-slave1'"
and check to make sure that only the key(s) you wanted were added.
```

```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-slave2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/
.ssh/id_rsa.pub"
The authenticity of host 'hadoop-slave2 (172.21.173.133)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kSO/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-slave2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'hadoopuser@hadoop-slave2'"
and check to make sure that only the key(s) you wanted were added.
```

```
hadoopuser@hadoop-master:/home/hadoop$ ssh-copy-id hadoopuser@hadoop-slave3
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hadoopuser/
.ssh/id_rsa.pub"
The authenticity of host 'hadoop-slave3 (172.21.173.134)' can't be established.
ED25519 key fingerprint is SHA256:Hh79Lh9kSO/ArJM4+TZcCrRdex2dyFFt1Iz+Us0ljaI.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
hadoopuser@hadoop-slave3's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'hadoopuser@hadoop-slave3'"
and check to make sure that only the key(s) you wanted were added.
```

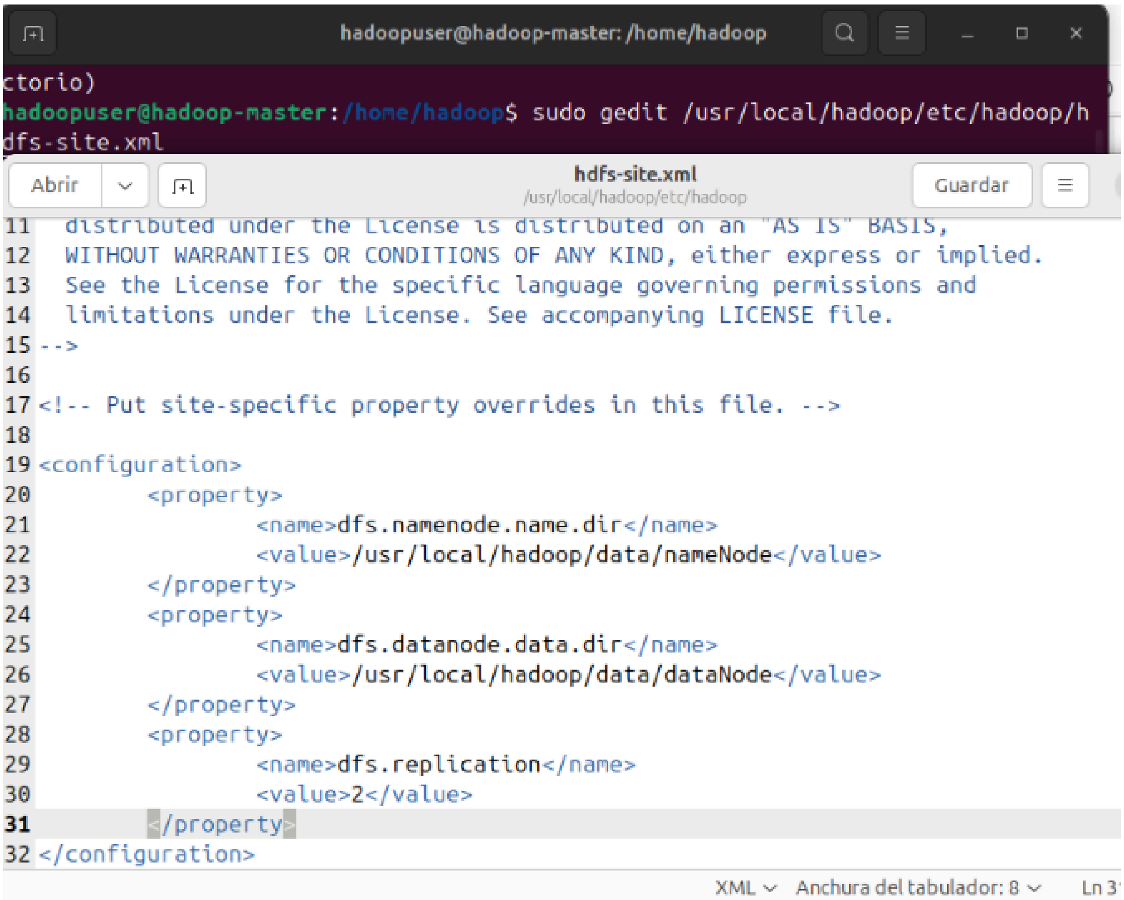
24. En el master, abrimos el archivo core-site.xml y añadimos lo siguiente:

```
<configuration>
        <property>
                <name>fs.defaultFS</name>
                <value>hdfs://hadoop-master:9000</value>
        </property>
</configuration>
```

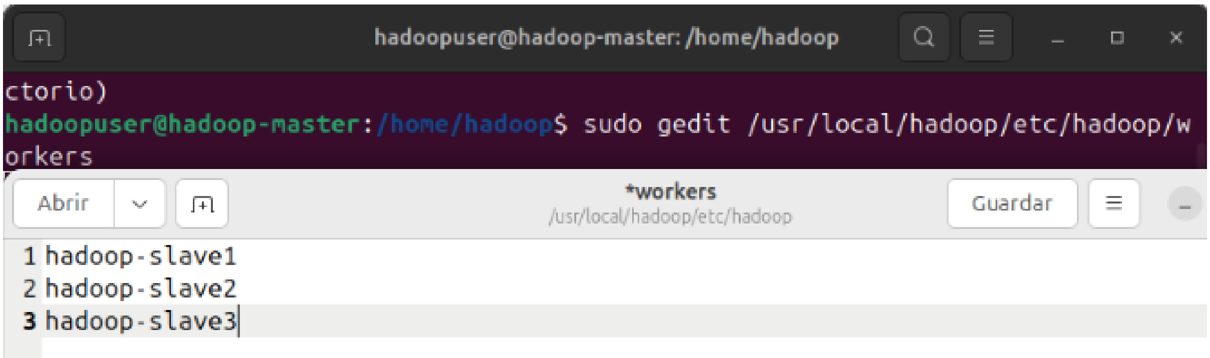


25. En el master, abrimos el archivo hdfs-site.xml

```
<configuration>
        <property>
                <name>dfs.namenode.name.dir</name>
                <value>/usr/local/hadoop/data/nameNode</value>
        </property>
        <property>
                <name>dfs.datanode.data.dir</name>
                <value>/usr/local/hadoop/data/dataNode</value>
        </property>
        <property>
                <name>dfs.replication</name>
                <value>2</value>
        </property>
</configuration>
```

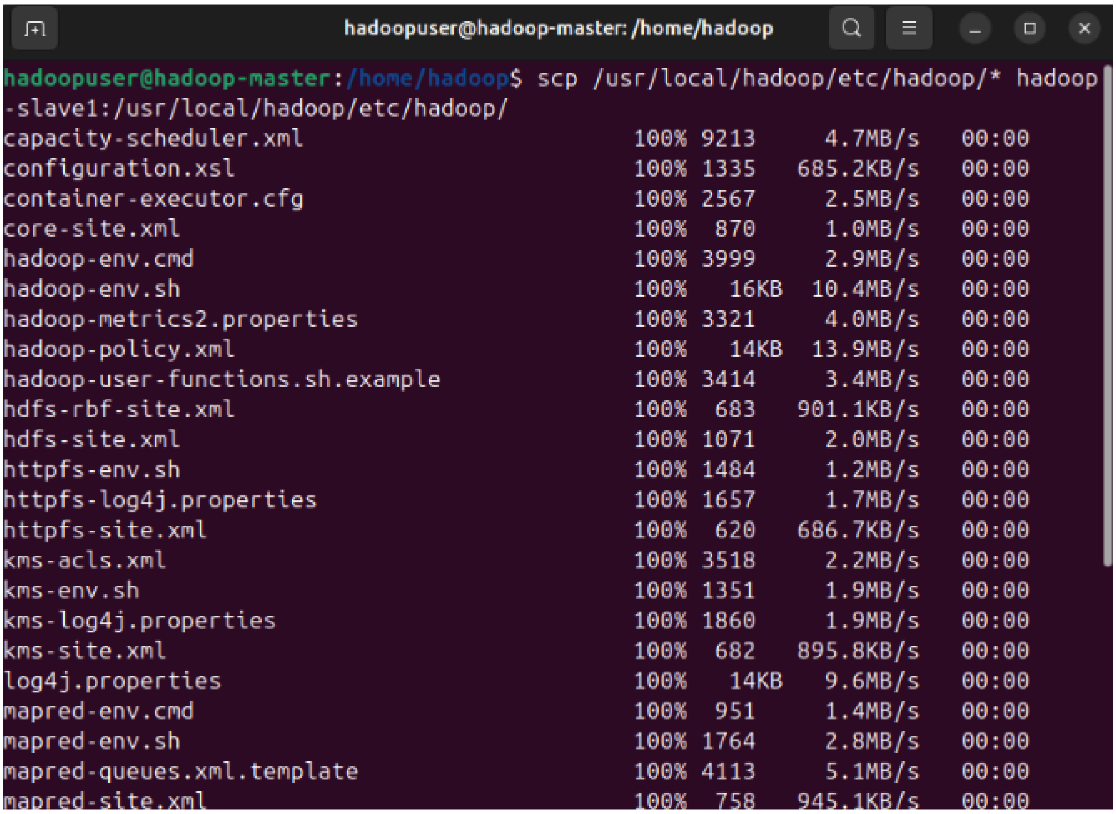


26. Abrimos el archivo workers y ponemos el nombre de los slaves.



27. Usamos los siguientes comandos para copiar la configuración del master a los slaves (un comando por cada slave):

scp /usr/local/hadoop/etc/hadoop/*
hadoop-slave1:/usr/local/hadoop/etc/hadoop/



```
hadoopuser@hadoop-master: /home/hadoop

hadoopuser@hadoop-master:/home/hadoop$ scp /usr/local/hadoop/etc/hadoop/* hadoop
-slave2:/usr/local/hadoop/etc/hadoop/
capacity-scheduler.xml                        100% 9213     1.2MB/s   00:00
configuration.xsl                             100% 1335   216.5KB/s   00:00
container-executor.cfg                        100% 2567   388.4KB/s   00:00
core-site.xml                                 100%  870   123.1KB/s   00:00
hadoop-env.cmd                                100% 3999   420.3KB/s   00:00
hadoop-env.sh                                 100%   16KB   9.8MB/s   00:00
hadoop-metrics2.properties                    100% 3321     4.2MB/s   00:00
hadoop-policy.xml                             100%   14KB  12.8MB/s   00:00
hadoop-user-functions.sh.example              100% 3414   454.3KB/s   00:00
hdfs-rbf-site.xml                             100%  683   473.0KB/s   00:00
hdfs-site.xml                                 100% 1071   954.4KB/s   00:00
httpfs-env.sh                                 100% 1484    82.0KB/s   00:00
httpfs-log4j.properties                       100% 1657     2.5MB/s   00:00
httpfs-site.xml                               100%  620     1.0MB/s   00:00
kms-acls.xml                                  100% 3518   354.5KB/s   00:00
kms-env.sh                                    100% 1351     2.0MB/s   00:00
kms-log4j.properties                          100% 1860     2.5MB/s   00:00
kms-site.xml                                  100%  682   956.8KB/s   00:00
log4j.properties                              100%   14KB  16.1MB/s   00:00
mapred-env.cmd                                100%  951     1.5MB/s   00:00
mapred-env.sh                                 100% 1764     3.3MB/s   00:00
mapred-queues.xml.template                    100% 4113     5.5MB/s   00:00
mapred-site.xml                               100%  758   941.5KB/s   00:00
```

```
hadoopuser@hadoop-master: /home/hadoop

hadoopuser@hadoop-master:/home/hadoop$ scp /usr/local/hadoop/etc/hadoop/* hadoop
-slave3:/usr/local/hadoop/etc/hadoop/
capacity-scheduler.xml                        100% 9213     3.8MB/s   00:00
configuration.xsl                             100% 1335   960.2KB/s   00:00
container-executor.cfg                        100% 2567     2.5MB/s   00:00
core-site.xml                                 100%  870   953.4KB/s   00:00
hadoop-env.cmd                                100% 3999     4.7MB/s   00:00
hadoop-env.sh                                 100%   16KB   8.7MB/s   00:00
hadoop-metrics2.properties                    100% 3321     5.2MB/s   00:00
hadoop-policy.xml                             100%   14KB  18.8MB/s   00:00
hadoop-user-functions.sh.example              100% 3414     3.7MB/s   00:00
hdfs-rbf-site.xml                             100%  683     1.4MB/s   00:00
hdfs-site.xml                                 100% 1071     2.2MB/s   00:00
httpfs-env.sh                                 100% 1484     2.1MB/s   00:00
httpfs-log4j.properties                       100% 1657     2.9MB/s   00:00
httpfs-site.xml                               100%  620     1.2MB/s   00:00
kms-acls.xml                                  100% 3518     4.6MB/s   00:00
kms-env.sh                                    100% 1351     1.8MB/s   00:00
kms-log4j.properties                          100% 1860     3.2MB/s   00:00
kms-site.xml                                  100%  682     1.2MB/s   00:00
log4j.properties                              100%   14KB  14.8MB/s   00:00
mapred-env.cmd                                100%  951     1.5MB/s   00:00
mapred-env.sh                                 100% 1764     2.6MB/s   00:00
mapred-queues.xml.template                    100% 4113     2.0MB/s   00:00
mapred-site.xml                               100%  758     1.3MB/s   00:00
```

28. Formateamos el sistema de archivos de HDFS.

```
hadoopuser@hadoop-master:~$ source /etc/environment
hadoopuser@hadoop-master:~$ hdfs namenode -format
WARNING: /usr/local/hadoop/logs does not exist. Creating.
2025-11-06 19:58:46,215 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = hadoop-master/172.21.173.135
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.4.2
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/
hadoop/common/lib/jetty-server-9.4.57.v20241219.jar:/usr/local/hadoop/share/hado
op/common/lib/netty-resolver-dns-native-macos-4.1.118.Final-osx-aarch_64.jar:/us
r/local/hadoop/share/hadoop/common/lib/netty-buffer-4.1.118.Final.jar:/usr/local
/hadoop/share/hadoop/common/lib/jsp-api-2.1.jar:/usr/local/hadoop/share/hadoop/c
ommon/lib/commons-io-2.16.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-
servlet-9.4.57.v20241219.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-tra
nsport-classes-kqueue-4.1.118.Final.jar:/usr/local/hadoop/share/hadoop/common/li
b/commons-daemon-1.0.13.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-tran
sport-native-epoll-4.1.118.Final-linux-riscv64.jar:/usr/local/hadoop/share/hadoo
p/common/lib/kerby-asn1-2.0.3.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb
-crypto-2.0.3.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-xml-9.4.57.v20
241219.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-http-9.4.57.v20241219
```

29. Encendemos HDFS.

```
hadoopuser@hadoop-master:~$ start-dfs.sh
Starting namenodes on [hadoop-master]
Starting datanodes
hadoop-slave3: WARNING: /usr/local/hadoop/logs does not exist. Creating.
hadoop-slave1: WARNING: /usr/local/hadoop/logs does not exist. Creating.
hadoop-slave2: WARNING: /usr/local/hadoop/logs does not exist. Creating.
Starting secondary namenodes [hadoop-master]
```

Comprobamos si se ha encendido correctamente.

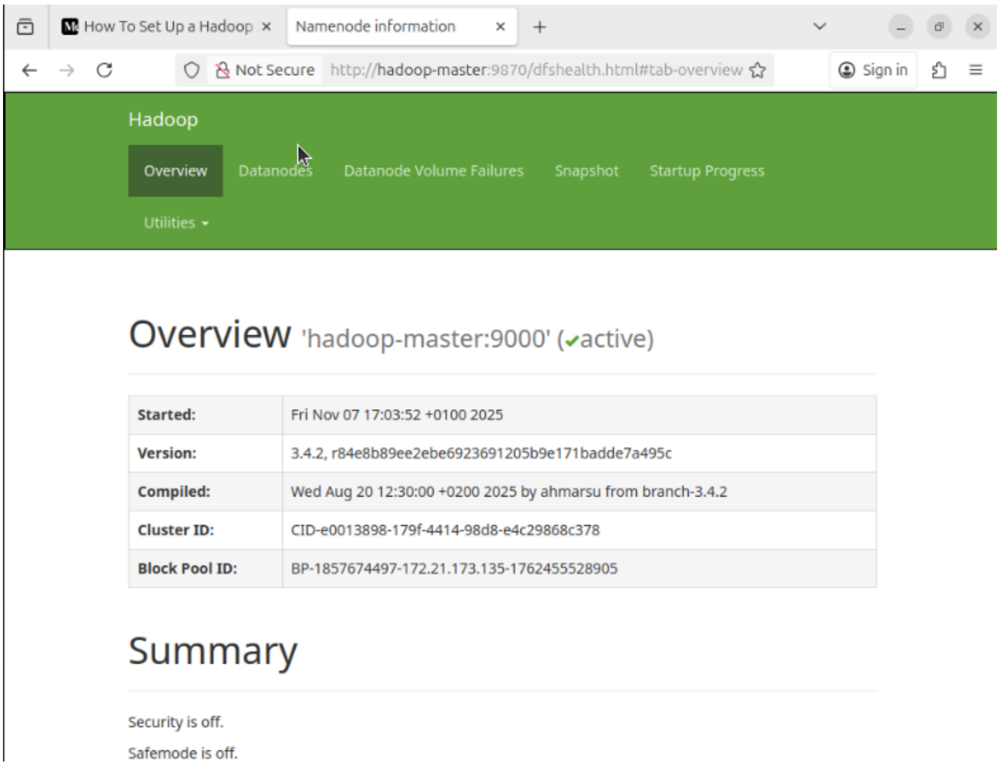
```
hadoopuser@hadoop-master:~$ jps
5907 SecondaryNameNode
6039 Jps
5655 NameNode
```

```
hadoop@hadoop-slave1:~$ jps
5420 Jps
```

```
hadoop@hadoop-slave2:~$ jps
5903 Jps
```

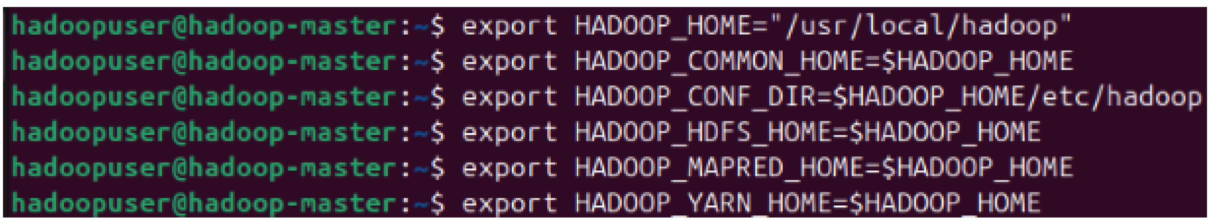
```
hadoop@hadoop-slave3:~$ jps
5473 Jps
```

30. Ponemos en el navegador hadoop-master:9870



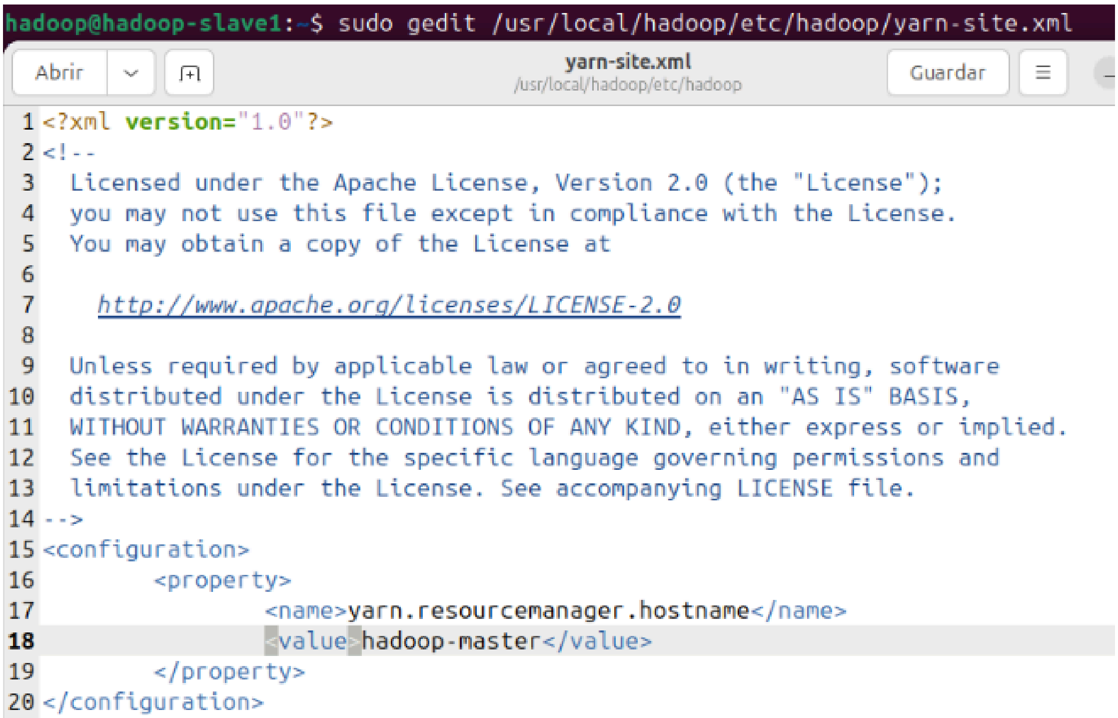
31. Configuramos yarn con los siguientes comandos:

export HADOOP_HOME="/usr/local/hadoop"
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export HADOOP_HDFS_HOME=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_YARN_HOME=$HADOOP_HOME



32. En los slaves abrimos el archivo yarn-site.xml y añadimos lo siguiente:

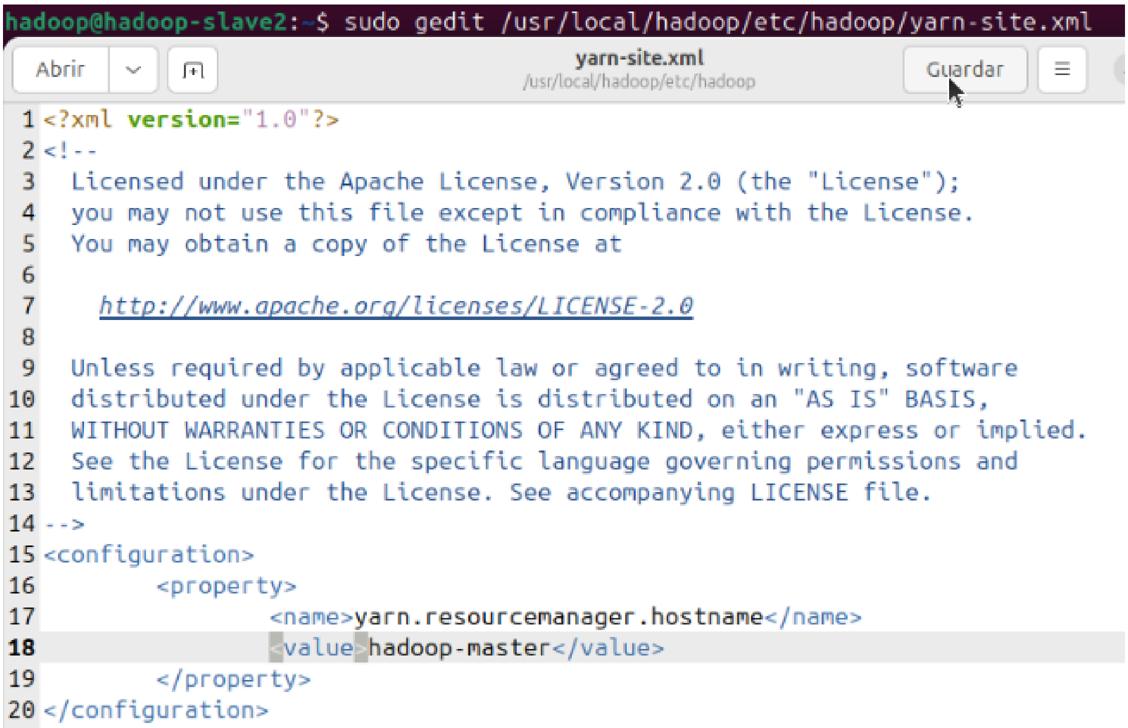
<property>
<name>yarn.resourcemanager.hostname</name>
<value>hadoop-master</value>
</property>



```
hadoop@hadoop-slave1:~$ sudo gedit /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

Abrir | yarn-site.xml /usr/local/hadoop/etc/hadoop | Guardar

```xml
<?xml version="1.0"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>
        <property>
                <name>yarn.resourcemanager.hostname</name>
                <value>hadoop-master</value>
        </property>
</configuration>
```



```
hadoop@hadoop-slave2:~$ sudo gedit /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

Abrir | yarn-site.xml /usr/local/hadoop/etc/hadoop | Guardar

```xml
<?xml version="1.0"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>
        <property>
                <name>yarn.resourcemanager.hostname</name>
                <value>hadoop-master</value>
        </property>
</configuration>
```

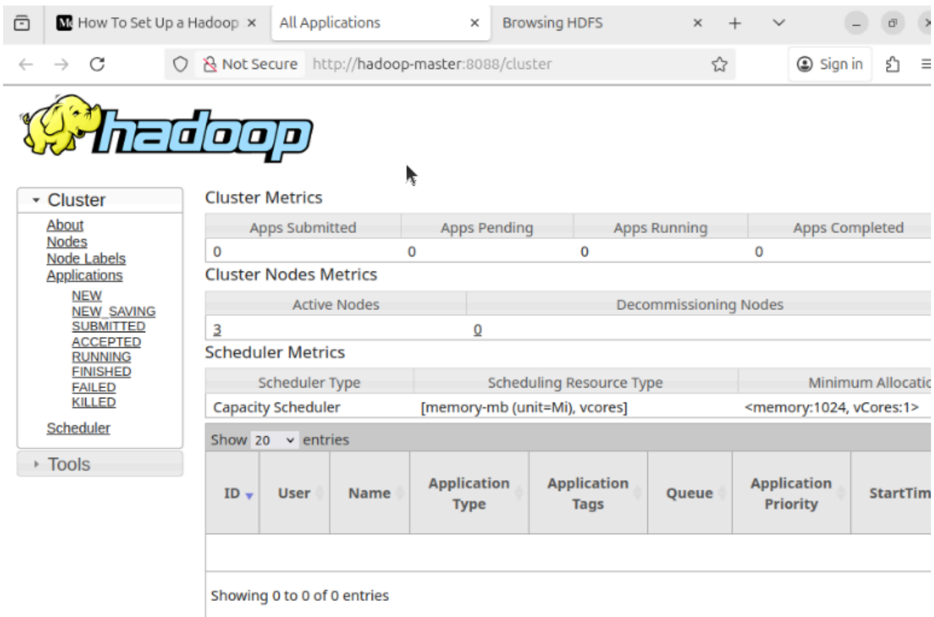
```
hadoop@hadoop-slave3:~$ sudo gedit /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

```
<?xml version="1.0"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>
        <property>
                <name>yarn.resourcemanager.hostname</name>
                <value>hadoop-master</value>
        </property>
</configuration>
```

33. Encendemos yarn.

```
hadoopuser@hadoop-master:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

34. Escribimos en el navegador: http://hadoop-master:8088/cluster



Conclusiones

Mis conclusiones son que instalar hadoop es un trabajo con muchos pasos que hay que seguir al pie de la letra para que no haya ningún error. Hay que tener claro cuáles son las IPs de cada máquina y separarlas bien. Por lo demás, mientras se sigan bien los pasos, no es una instalación complicada.

Problemas encontrados/soluciones

En mi caso no he tenido ningún problema, he seguido cada paso de la instalación y ha funcionado todo correctamente.